Marc Jourdan

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Academic and industry positions

Academic positions

2024 – Present **Postdoctoral researcher**, TML Laboratory, **EPFL**, Lausanne, Switzerland.
Study learning theory for large language models, focused on post-training.
Advisor: Prof. Dr. Nicolas Flammarion.

2021 – 2024 **Graduate researcher**, Inria Scool (CRIStAL), **Université de Lille**, Lille, France.
Studied pure exploration problems with the Top Two approach.
Advisors: Dr. Émilie Kaufmann and Dr. Rémy Degenne.

Visiting positions

2024 (3 mo.) **Visiting researcher**, LAILA, **Università degli Studi di Milano**, Milan, Italy.
Studied regret minimization for smoothed adversarial linear contextual bandits.
Advisor: Prof. Dr. Nicolò Cesa-Bianchi.

2021 (5 mo.) **Visiting researcher**, Inria Scool (CRIStAL), **Université de Lille**, Lille, France.
Studied bandit identification with continuous answers.
Advisor: Dr. Rémy Degenne.

2020 (6 mo.) **Visiting researcher**, Learning & Adaptive Systems, **ETH Zürich**, Zürich, Switzerland.
Studied bandit identification with combinatorial arms.
Advisors: Dr. Mojmír Mutný, Dr. Johannes Kirschner and Prof. Dr. Andreas Krause.

Industry positions

2019 (6 mo.) **Data scientist** (part-time), **AMAG Leasing**, Zürich, Switzerland.
Created a recommender system for customers, developed churn prediction models.

2018 (5 mo.) **Research intern**, AI @ Nation Scale, **IBM Research**, Singapore.
Characterized entities in the Bitcoin blockchain, probabilistically modeled its evolution.
Advisors: Dr. Sébastien Blandin, Dr. Laura Wynter and Dr. Pralhad Deshpande.

2017 (3 mo.) **Research intern**, **ST Microelectronics**, Crolles, France.
Quantized convolutional neural network for electronic chip.
Advisor: Pascal Urard.

Education

2021 – 2024 **Ph.D. in Computer Science**, Inria Scool (CRIStAL), **Université de Lille**, Lille, France.
Multi-Armed Bandits, Pure Exploration, Best Arm Identification, Differential Privacy.
Thesis title: *Solving Pure Exploration Problems with the Top Two Approach.*
Advisors: Dr. Émilie Kaufmann (CNRS) and Dr. Rémy Degenne (Inria).
Committee: Prof. Dr. Wouter Koolen (Chair), Prof. Dr. Aurélien Garivier (Examiner), Prof. Dr. Alexandre Proutière and Prof. Dr. Sandeep Juneja (External Reviewers).
Thesis manuscript: `https://theses.fr/2024ULILB011`.

2018 – 2020 **M.Sc. Data Science**, *with distinction* (GPA 5.8/6), **ETH Zürich**, Zürich, Switzerland.
Statistics, Machine Learning. Master's Thesis in the *Learning & Adaptive Systems* group.
Thesis title: *Pure Exploration for Combinatorial Bandits with Semi-Bandit Feedback.*
Advisors: Dr. Mojmír Mutný, Dr. Johannes Kirschner and Prof. Dr. Andreas Krause.
Thesis manuscript: `http://hdl.handle.net/20.500.11850/523561`

2015 – 2019 **Ingénieur (M.Sc.)**, *with distinction* (GPA 3.92/4), **École Polytechnique**, Palaiseau, France.
Applied Mathematics, Computer Science.

Education (continued)

2013 – 2015 **Classes préparatoires** MPSI/MP*, **Lycée Louis-Le-Grand**, Paris, France.
Mathematics, Physics, Computer Science.

Awards, fellowships and grants

Awards

2025 Runner-up (ex aequo) for the **PhD Award in Machine Learning** of the SSFAM (Francophone Learned Society for Machine Learning).

Runner-up (ex aequo) for the **PhD Award in Artificial Intelligence** of the AFIA (French Association for Artificial Intelligence).

Fellowships

2021 – 2024 **Doctoral fellowship AI_PhD@Lille** (3 years) at the University of Lille in the *Inria Scool* team with Dr. Émilie Kaufmann and Dr. Rémy Degenne, THIA ANR program.

Grants

2024 **Mobility grant** (3 months) at the University of Milan in the *Laboratory of Artificial Intelligence and Learning Algorithms* with Prof. Dr. Nicolò Cesa-Bianchi, Program "France 2030" (SFRI project GRAEL), 2250 €.

Publications

Ordering policy: Authors are ordered by contribution; * denotes equal contribution, advisors are listed last.
Summary statistics: journals (2 JMLR), conferences (5 NeurIPS, 2 ICML, 2 AISTATS, 2 ALT), workshops (3).

International journals

[AJAB26] A. Azize*, M. **Jourdan***, A. Al Marjani, and D. Basu, "Differentially private best-arm identification," *Journal of Machine Learning Research (JMLR)*, 2026.

[JDR26] M. **Jourdan**, A. Delahaye-Duriez, and C. Réda, "An anytime algorithm for good arm identification," *Journal of Machine Learning Research (JMLR)*, 2026.

International conferences with proceedings

[JYF25] M. **Jourdan**, G. Yüce, and N. Flammarion, "Learning parametric distributions from samples and preferences," *International Conference on Machine Learning (ICML)*, 2025, **Spotlight** (2.6%).

[JA25] M. **Jourdan*** and A. Azize*, "Optimal best arm identification under differential privacy," *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[KJK25] C. Kone, M. **Jourdan**, and E. Kaufmann, "Pareto set identification with posterior sampling," *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.

[PJDK25] R. Poiani*, M. **Jourdan***, R. Degenne, and E. Kaufmann, "Best-arm identification in unimodal bandits," *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.

[AJAB23] A. Azize, M. **Jourdan**, A. Al Marjani, and D. Basu, "On the complexity of differentially private best-arm identification with fixed confidence," *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[JD23] M. **Jourdan** and R. Degenne, "Non-asymptotic analysis of a ucb-based top two algorithm," *Advances in Neural Information Processing Systems (NeurIPS)*, 2023, **Spotlight** (3.06%).

[JDK23a] M. **Jourdan**, R. Degenne, and E. Kaufmann, "An ε-best-arm identification algorithm for fixed-confidence and beyond," *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[JDK23b] M. **Jourdan**, R. Degenne, and E. Kaufmann, “Dealing with unknown variances in best-arm identification,” *Algorithmic Learning Theory (ALT)*, 2023.

[JD22] M. **Jourdan** and R. Degenne, “Choosing answers in ε-best-answer identification for linear bandits,” *International Conference on Machine Learning (ICML)*, 2022.

[JDB+22] M. **Jourdan**, R. Degenne, D. Baudry, R. De Heide, and E. Kaufmann, “Top two algorithms revisited,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[JMKK21] M. **Jourdan**, M. Mutný, J. Kirschner, and A. Krause, “Efficient pure exploration for combinatorial bandits with semi-bandit feedback,” *Algorithmic Learning Theory (ALT)*, 2021.

International workshops with proceedings

[JMRS21] M. **Jourdan**, K. Martinkus, D. Roschewitz, and M. Strohmeier, “I know where you are going: Predicting flight destinations of corporate and state aircraft,” in *Engineering Proceedings*, 2021.

[JBWD19] M. **Jourdan**, S. Blandin, L. Wynter, and P. Deshpande, “A probabilistic model of the bitcoin blockchain,” in *Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019.

[JBWD18] M. **Jourdan**, S. Blandin, L. Wynter, and P. Deshpande, “Characterizing entities in the bitcoin blockchain,” in *International Conference on Data Mining Workshops (ICDMW)*, 2018.

Preprints

[JF26] M. **Jourdan** and N. Flammarion, “Learning distributions from samples and qualities,” 2026.

Invited talks

Seminar talks

2026 **CMAP**, École Polytechnique. March 18th.
Advances in Pure Exploration in Bandits: Non-Asymptotic and Private.

2025 **CREST**, ENSAE. November 10th.
Advances in Pure Exploration in Bandits: Non-Asymptotic and Private.

Ghost, Inria Grenoble. October 23rd.
Advances in Pure Exploration in Bandits: Non-Asymptotic, Private, and Structured.

Malt, Inria Rennes. October 9th.
Advances in Pure Exploration in Bandits: Non-Asymptotic, Private, and Structured.

ILOCOS, CentraleSupélec. October 7th.
Advances in Pure Exploration in Bandits: Non-Asymptotic, Private, and Structured.

ML-MTP, Inria Montpellier. October 2nd.
Advances in Pure Exploration in Bandits: Non-Asymptotic, Private, and Structured.

2024 **LAILA**, University of Milan. June 11th.
Solving pure exploration problems with the Top Two approach.

FLAIR, EPFL. February 19th.
Solving pure exploration problems with the Top Two approach.

Data Science, University of Neuchâtel. February 13th.
Solving pure exploration problems with the Top Two approach.

2023 **LAS**, ETH Zürich. February 1st.
Best-Arm Identification with Top Two Algorithms.

2022 **Scool**, Inria Lille. June 17th.
Top Two Algorithms Revisited.

Invited talks (continued)

2021 **Scool**, Inria Lille. November 19th.
Choosing Answers in ε-Best-Answer Identification for Linear Bandits.

2020 **LAS**, ETH Zürich. October 14th.
Efficient Pure Exploration for Combinatorial Bandits with Semi-Bandit Feedback.

Conference talks

2026 **PEPR IA Days**, Rennes. June 23rd.
Advances in Private Pure Exploration in Bandits.

2025 **CAp**, Dijon. July 1st.
Runner-up PhD Award SSFAM: Solving pure exploration problems with the Top Two approach.

2024 **CAp**, Lille. July 1st.
Differentially private best-arm identification.

2023 **NeurIPS@Paris**, Paris. December 7th.
Non-Asymptotic Analysis of a UCB-based Top Two Algorithm.
An ε-Best-Arm Identification Algorithm for Fixed-Confidence and Beyond.

ALT, Singapore. February 21st.
Dealing with unknown variances in best-arm identification.

2022 **NeurIPS@Paris**, Paris. November 23rd.
Top Two Algorithms Revisited.

StatMathAppli, Fréjus. September 1st.
Top Two Algorithms Revisited.

ICML, Baltimore. July 20th.
Choosing Answers in ε-Best-Answer Identification for Linear Bandits.

2021 **ALT**, Paris. March 9th (virtual).
Efficient Pure Exploration for Combinatorial Bandits with Semi-Bandit Feedback.

Teaching and supervision

Teaching

2022 **Teaching assistant**, *Computational Statistics*, M.Sc. (1st year), tutorials (24h), $\sim$ 30 students, **Université de Lille**, Lille, France.

2020 **Teaching assistant**, *Machine Perception*, M.Sc. (1st year), tutorials (18h), $\sim$ 60 students, **ETH Zürich**, Zürich, Switzerland.

Supervision

2025 **Maryna Horbach**, summer intern before M.Sc. at Oxford, TML Laboratory, EPFL.
Topic: Elimination strategies for fixed-budget best-arm identification.

Gizem Yüce, PhD student, TML Laboratory, EPFL.
Topic: Learning parametric distributions from samples and preferences (ICML 2025).

Referee activities

2025 – 2026 International Conference on Neural Information Processing Systems (NeurIPS).

2024 – 2025 International Conference on Machine Learning (ICML).

2023 – 2026 International Conference on Artificial Intelligence and Statistics (AISTATS).

2021 & 2026 International Conference on Algorithmic Learning Theory (ALT).

2023 – 2026 European Workshop on Reinforcement Learning (EWRL).

Referee activities (continued)

2022 IEEE Journal on Selected Areas in Information Theory (JSAIT).